

TF-IDF v. MLP: Fake news detection with computational models

De Angeli Pietro, Mnev Vadim, Vanto Otto

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1 Abstract

The rise of fake news poses a critical threat to public trust, democratic stability, and information integrity. In this study, we develop and compare two computational approaches for fake news detection using natural language processing techniques. Using a labeled dataset of over 44000 news articles, we evaluate both a term frequency–inverse document frequency (TF-IDF) model and a more complex multilayer perceptron (MLP) classifier trained on semantic embeddings generated via the OpenAI API. The TF-IDF model provides interpretable insights into key word-level indicators of misinformation (with an accuracy of 98.17 %), while the MLP achieves higher classification accuracy (99.10 %). Text analysis reveals that fake news articles often rely on emotionally charged and speculative language, whereas real news tends to use more formal and fact-based vocabulary. Our findings highlight the trade-off between model interpretability and accuracy, and offer practical guidance for deploying automated fake news detection tools in real-world scenarios.

2 Introduction

In recent years, detecting fake news has emerged as a challenge in the digital era, influencing everything from public opinion to election outcomes and public health behavior (Shu, et al., 2017; Lazer, et al., 2018). The rapid spread of misinformation on social media platforms and online news outlets undermines trust in institutions, polarizes communities and spreads harmful narratives (Zhou & Zafarani, 2020). Therefore, developing effective tools for automatic fake news detection has become a priority for researchers, journalists, and policymakers (Ahmed, et al., 2017).

Fake news detection is a complex task that includes many disciplines, including natural language processing, machine learning, and social science. Fake news often tries to mimic the structure and tone of legitimate journalism, making it difficult to detect (Lazer, et al., 2018). It requires analyzing both linguistic features and context, such as source credibility and publication patterns.

In this project, we address the problem of fake news detection through a text-based machine learning approach. We apply modern techniques to a large, labeled dataset of news articles and evaluate the performance of different supervised learning models. Our goal is to build two models that detect fake news among news articles only by the contents of the article, and compare these models and their results and draw conclusions about the capability of detecting fake news and its caveats in real world situations.

To this end, we pre-process the dataset to standardize and clean a dataset containing over 44000 news articles, normalize inconsistent subject labels and create semantic embeddings of the articles using OpenAI API. These embeddings are then classified using a multilayer perceptron (MLP), a type of neural network suited for supervised learning tasks (Zhang et al., 2015) and term frequency–inverse document frequency (TF-IDF), widely used in misinformation detection (Ahmed et al., 2017; Zhou & Zafarani, 2020). In addition to evaluating model performance, we also perform qualitative analysis of the language used in fake versus real news to gain insights into stylistic and rhetorical differences (Horne & Adali, 2017). We then discuss the implications of the differences between these models and their results.

3 Dataset

To develop our fake news classification model, we used a publicly available dataset that contains labeled news articles categorized as either fake or real. The dataset, titled the *Fake and Real News Dataset*, consists of two separate groups: 23,502 articles labeled as fake news and 21,417 labeled as real news. Each article includes several metadata fields: title, body text, subject, publication date, and a binary label indicating authenticity.

The dataset was obtained from Hugging Face and is accessible at: <https://huggingface.co/>

To prepare the data for modeling, we implemented a pre-processing pipeline. This included merging the `title` and `body text` fields to create a unified textual input for each article. We also applied standard text-cleaning steps: converting all text to lowercase, removing punctuation, and eliminating stop words. These transformations help reduce noise and ensure consistency in the input data.

Example before pre-processing:

- Title: Brazil prosecutors seek to extend Batista detention, source says
- Body text: The Office of Brazil s Prosecutor-General Rodrigo Janot asked the Federal Supreme Court on Thursday to convert the temporary detention of billionaire beef tycoon Joesley Batista into preventative detention, a person with knowledge of the situation said.
- Subject: worldnews
- Publication Date: September 14, 2017
- Label: 1

After pre-processing:

- Merged text: brazil prosecutors seek extend batista detention source says office brazil prosecutor general rodrigo janot asked federal supreme court thursday convert temporary detention billionaire beef tycoon joesley batista preventative detention person knowledge situation said
- Subject: worldnews
- Publication date: September 14, 2017
- Label: 1

4 Topic normalization

We observed inconsistency in the dataset’s subject labels, which posed a challenge for any topic-based analysis or filtering. Labels such as "news," "worldnews," and "Middle-east" often overlapped semantically, while categories like "Left" or "Government" were disproportionately associated with either fake or real news. This imbalance risked introducing unintended biases during training.

After several iterations of testing and cleaning, we identified the following labels:

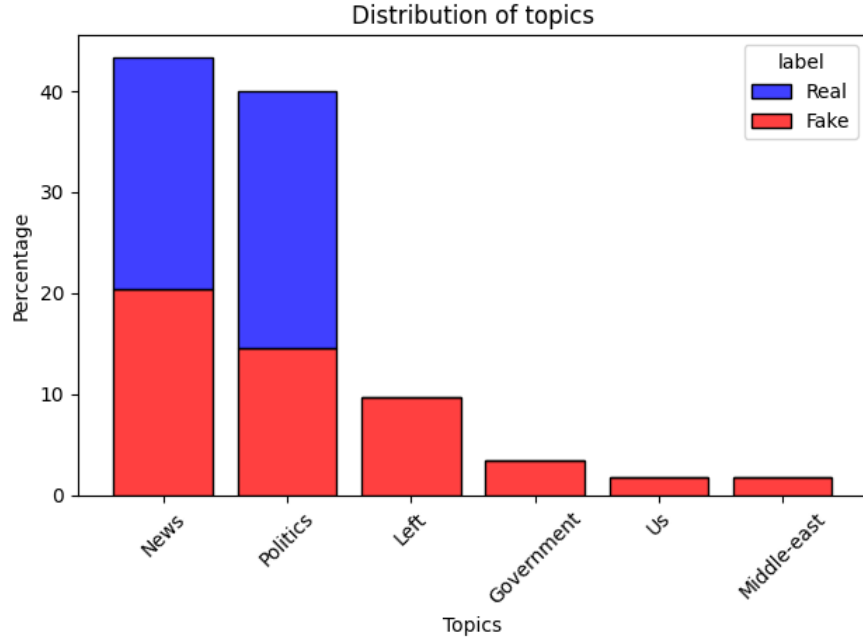


Figure 1: Original topic distribution

The most common subject labels in the dataset were:

- Politics
- News
- Left
- Government
- Middle-east
- US

To address this, we normalized the subject categories using a keyword-based matching approach. We consolidated the most frequently occurring subject labels into two broader and semantically coherent categories:

- **Politics:** includes original labels such as *Politics*, *Left*, and *Government*
- **News:** includes labels such as *News*, *Middle-east*, and *US*

This grouping reduced label noise, simplified the classification task, and helped ensure that the model was not influenced by underrepresented or skewed subject labels.

After merging the topics, the distribution became more balanced, as shown below.

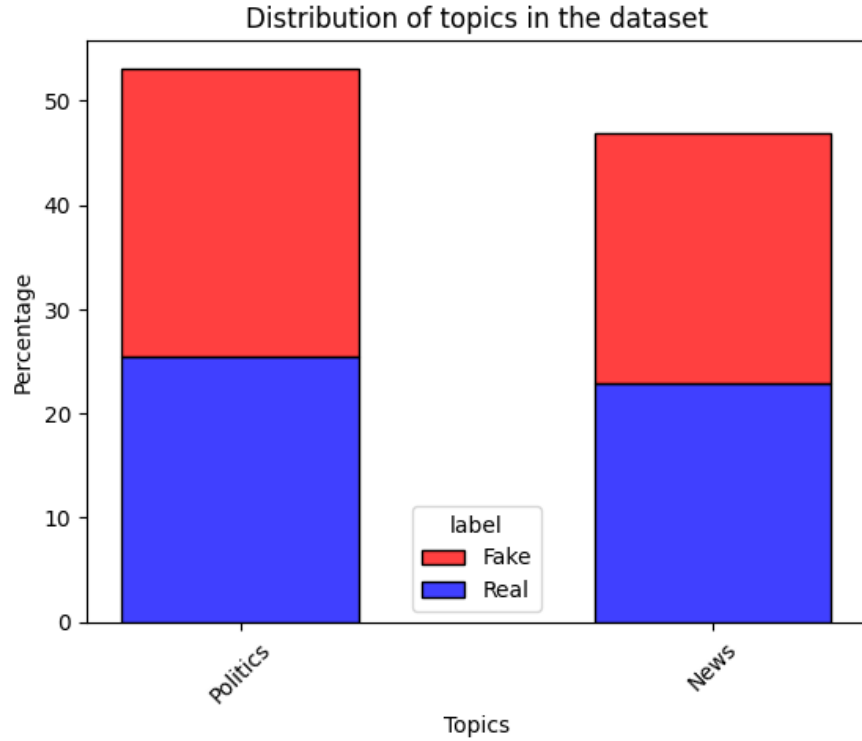


Figure 2: Balanced topic distribution after grouping

5 Embeddings

To accurately classify fake and real news articles based on their textual content, we first converted text into semantic embeddings. Given the relatively large size of our dataset (over 50,000 articles across training, validation, and test splits), generating embeddings using transformer-based models like BERT locally would have been computationally intensive. To overcome this, we utilized the OpenAI Embedding API, which allowed us to efficiently produce high-dimensional embeddings with minimal overhead.

Each article’s embedding was generated by concatenating its `title` and `body text` fields into a single string and submitting it to the API. The output for each article was a fixed-length vector of 1536 floating-point values, representing the semantic content of the article.

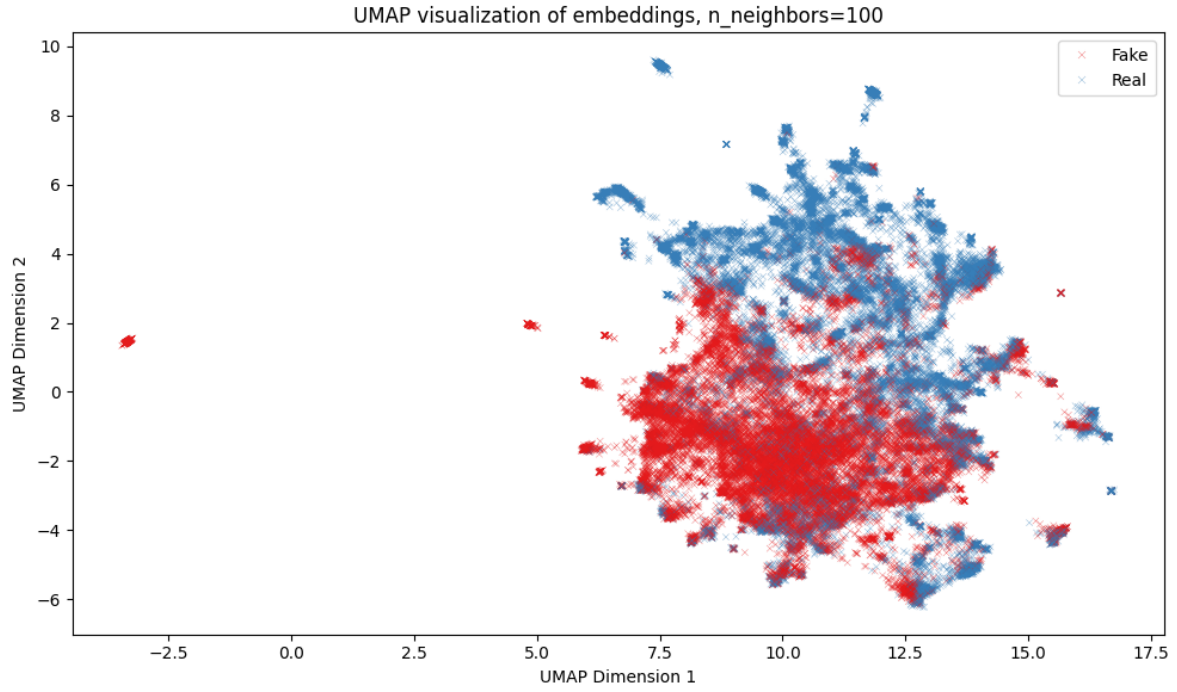


Figure 3: UMAP projection of the dataset embeddings (colored by label)

As shown in the plot, the classes are not linearly separable, but appear to be reasonably well distributed, with the average position of a group of embeddings of the two classes clearly distinguishable.

To further analyze the semantic structure, we colored the points based on the normalized topic labels:

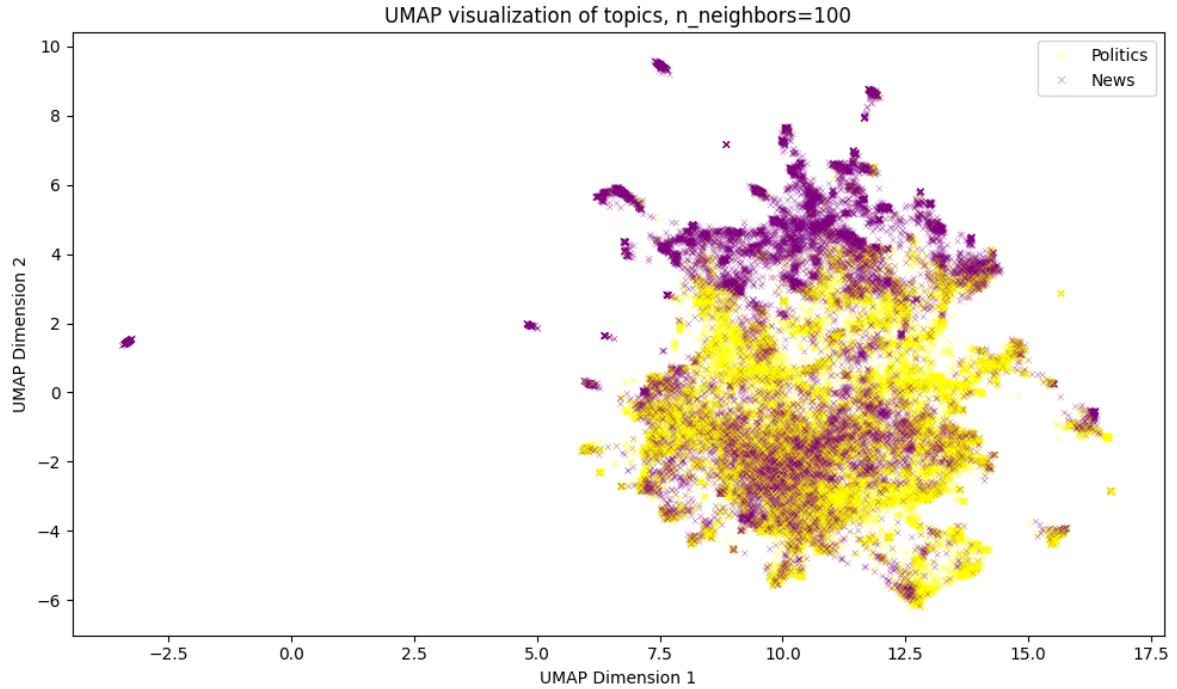


Figure 4: UMAP projection with topic coloring

We also visualized each topic category separately:

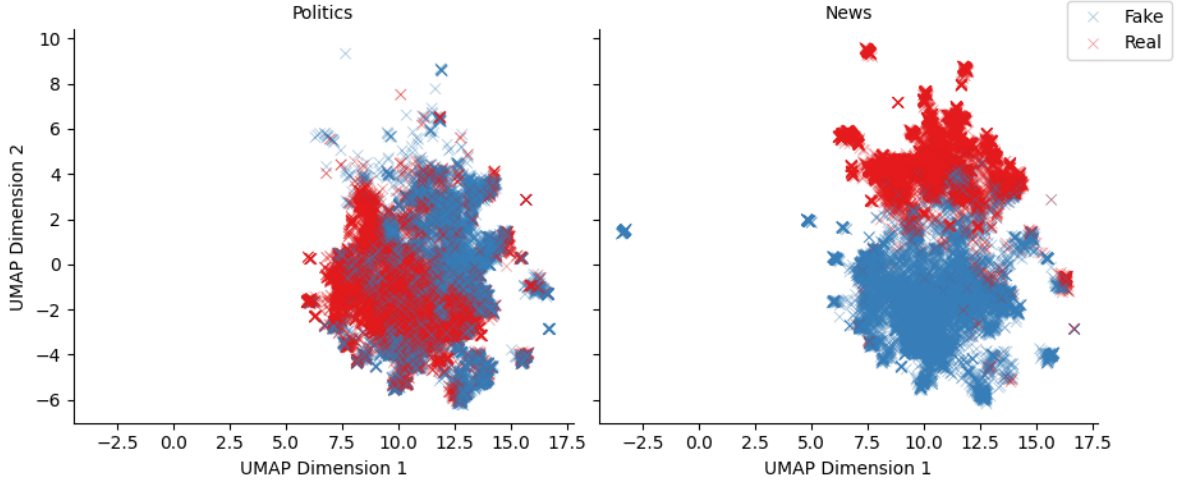


Figure 5: UMAP scatterplot per topic

Visualization of these embeddings revealed meaningful structure. In particular:

- Articles in the **News** category exhibited relatively clear separation between fake and real labels.
- Articles in the **Politics** category showed more overlap, suggesting that politically-themed fake news may be harder to distinguish due to subtle or misleading content that is semantically similar to real reporting.

6 Multilayer Perceptron (MLP)

We implemented a fully connected neural network using `scikit-learn`'s `MLPClassifier`. The network has two hidden layers with 256 and 128 neurons, respectively. The input layer size is 1536, matching the size of the embeddings. The output layer contains two neurons, corresponding to the two target classes. We used the validation set to enable early stopping and adopted the Adam optimizer for training.

7 Term frequency–inverse document frequency (TF-IDF)

We implemented TF-IDF on the embedded dataset to compare the results of two different models. We used `scikit-learn`'s `TfidfVectorizer` to fit the TF-IDF with the training dataset. We then predicted the fake or real news labels within the validation dataset.

8 Results

In this section we show the results of both models and discuss their characteristics.

8.1 MPL

The model achieved an accuracy of 99.10 % on the test set, indicating near-perfect performance.

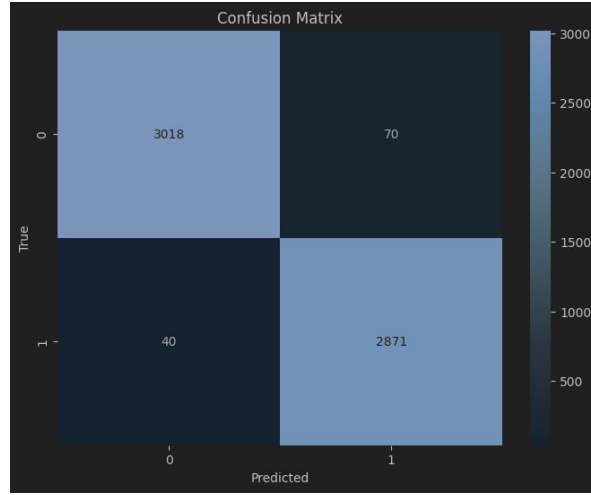


Figure 7: Confusion matrix of the test set

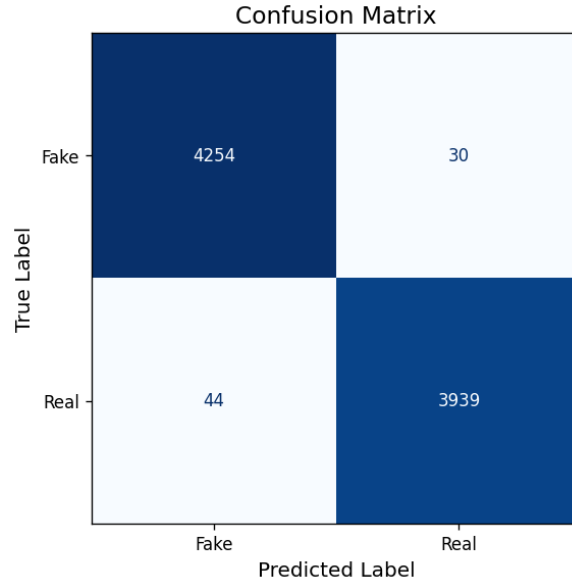


Figure 6: Confusion matrix on the test set

The difference between real news labeled as fake (0.011%) and fake news labeled as real (0.0071%) is only 0.004 percentage points. The confusion matrix is strongly diagonal, confirming that the model has minimal difficulty in distinguishing between fake and real news within the dataset.

8.2 TF-IDF

With TF-IDF we achieved 98.17 % accuracy.

The difference between real news labeled as fake (0.023%) and fake news labeled as real (0.013%) is 0.01 percentage points. The confusion matrix for TF-IDF is also strongly diagonal, confirming the model's accuracy in distinguishing between fake and real news, although TF-IDF was two times more prone to mislabel real news than fake news.

9 Text Analysis

We can analyze the text in the dataset, and try to show in some way the plots with a word cloud, then we can analyze the text.

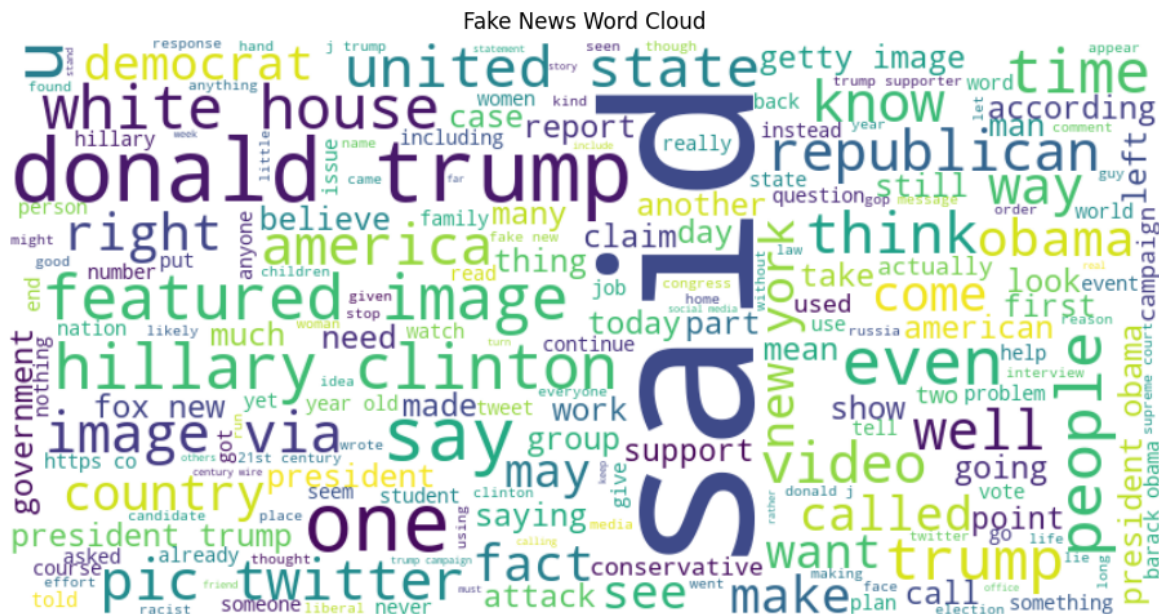


Figure 8: Word Cloud for Fake News

This word cloud represents the most frequent words in the fake news texts from the entire dataset. As we can see, many prominent words are associated with well-known public figures such as *Donald Trump*, *Obama*, and *Hillary Clinton*. We also observe the presence of subjective or speculative terms such as *think*, *believe*, *claim*, which are often paired with references to social media platforms like *Twitter*, *image*, and *video*. This suggests that fake news frequently relies on emotional or sensational language and tends to propagate quickly through social networks. The use of vague or opinion-based vocabulary might make these articles more attention-grabbing and persuasive for readers.

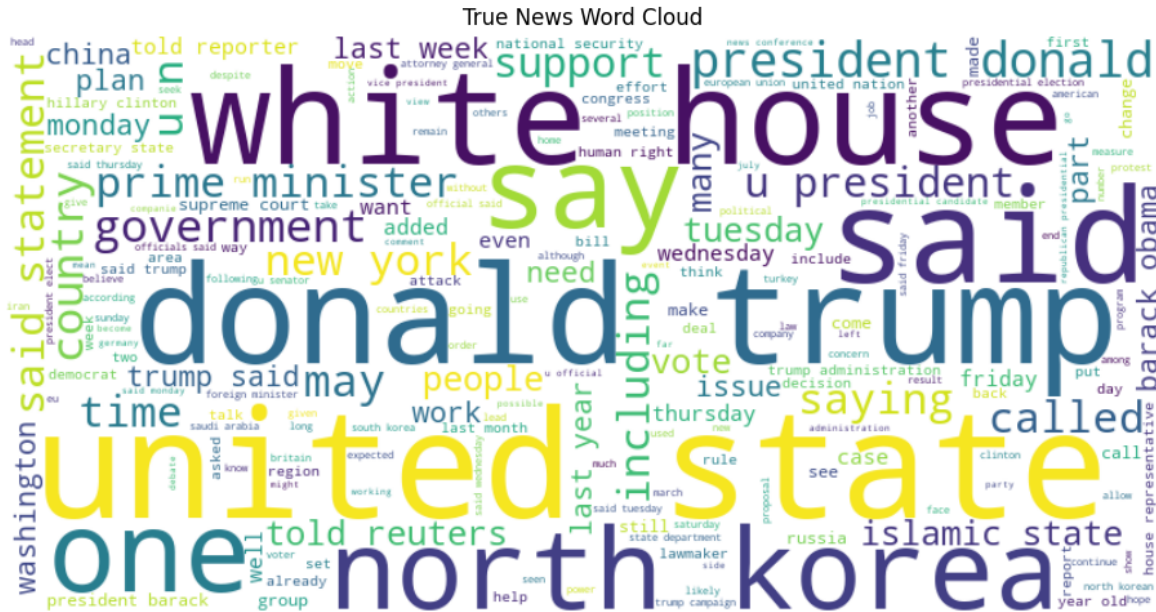


Figure 9: Word Cloud for True News

In contrast, the true news word cloud highlights similar references to political figures, but it also features more formal and informative vocabulary. Terms like *government*, *North Korea*, *statement*, *report*, and *official* suggest a focus on geopolitical topics and factual reporting. These elements reflect the typical style of professional journalism, which aims to present reliable, objective information. Compared to fake news, true news appears to use more precise language and references to credible sources and institutions.

10 Discussion

The differences between these models are most notably the time to develop the models and computational costs. TF-IDF is cheaper and quicker to implement and MLP is more time consuming and costly. TF-IDF also ignores the word order and context. In our case both of the models use embeddings which was the most costly part of our pipeline, so in this case the models are actually very similar. The TF-IDF could also have been used as the input for the MLP instead of a standalone predicting method, but not the other way around.

TF-IDF is less accurate (98.17 %) in predicting fake news articles, but by a small margin (2 pp). This is a very good result in the sense that detecting fake news is seemingly a straight-forward algorithmic problem that has only to do with the words used in the article. This is something that a human easily learn to do and therefore avoid misinformation. The tendency to mislabel real news almost twice as much as fake news is still problematic, while marginal in the context of a large dataset, could have significant implications if deployed at scale (Zhou & Zafarani, 2020).

MLP was more accurate (99.10 %) in predicting fake news articles, but the results are less understandable to humans. The neural network we developed was very capable of detecting fake and real news, but reveals no reasoning behind the labels (Doshi-Velez & Kim, 2017). This makes it more difficult to be certain of a models training and therefore motive for development. These kinds neural network models, while being capable of greater accuracy, can be used to spread misinformation because of the lack of perceivable reasoning behind the labeling.

In real world scenarios, especially in politics, the validity of the training set and the mindset of the readers in any given community changes the perception of fake news versus real news. Fake news as a term has been used by parties to obfuscate reality in order to drive the parties’ agenda forward. It is also trivial to teach these models with the intention of mislabeling certain kinds of news as fake news, so the integrity of the model is crucial in using a model to detect fake news (Shu et al., 2017; Zhou & Zafarani, 2020).

11 Conclusion

In this project, we explored two distinct approaches to fake news detection using natural language processing (TF-IDF) and supervised learning (MLP). The first model, TF-IDF, achieved 98.17% accuracy while offering transparent, interpretable insights into which terms most strongly signaled misinformation for the model. The second approach, MLP, demonstrated better accuracy at 99%, while seemingly capturing textual features and patterns that go beyond word frequency.

While the MLP model delivers higher predictive power, its black-box nature makes it less interpretable to humans and potentially more susceptible to bias or misuse, depending on the training set and application. TF-IDF, although marginally less accurate, offers clearer reasoning behind its classifications, which can be valuable in sensitive domains such as journalism or public policy.

Our analysis also revealed that fake news tends to use more emotional and vague language, often referencing well-known figures and speculative terms, while real news is characterized by more objective and structured reporting. Politically themed articles were found to be more challenging to classify, indicating the limitations of purely textual analysis in complex domains.

Ultimately, both models offer valuable perspectives: TF-IDF for human-understandable detection patterns, and MLP for high-performance classification. Future work could investigate hybrid approaches or incorporate contextual metadata (e.g., source reliability, user engagement) to improve robustness and explainability. As fake news continues to evolve, combining transparent, scalable, and accurate models will be key to maintaining information integrity in the digital age.

12 References

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